

## ANDREA PEREIRA

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### EDUCATION

#### Queen's University | Kingston, ON

*Expected Graduation, 2026*

Bachelors degree in Applied Science and Engineering

**Course Work:** Fundamentals of Information's Structures, Introduction to Data Science, Software Development Methodology, Introduction to Object Oriented Programming, Discrete Mathematics with Computer Engineering Applications, Computer Architecture, Image Processing and Computer Vision, Computer Vision and Deep Learning, Algorithms and Engineering Applications, Logic for Computing Science, Safety Critical Software, Operating Systems

### SKILLS & TECHNICAL TOOLS

**Languages:** Python, Java, C, C++, HTML, CSS, JavaScript

**Frameworks & Tools:** Flask, QGIS, TensorFlow, PyTorch, scikit-learn, NumPy, pandas, GitLab, MySQL, Android Studio, Qt Creator, Figma, VS Code, IntelliJ, GitHub, Linux, PyCharm

**Concepts:** Agile methodology, OOP, UI/UX Design, Graph Theory, Machine Learning, ISO 26262

### EXPERIENCE

#### Queen's Hyperloop Team | Kingston, ON

*Oct. 2024- Current*

- Engineered an optimized, cost-effective route from Toronto to Winnipeg by implementing Dijkstra's algorithm on geospatial road network data using Python, QGIS, and NetworkX
- Programmatically merged and clipped shapefiles from multiple provinces (Ontario and Manitoba), handling over 40,000 road segments to construct a continuous transportation network
- Built a weighted directed graph of real-world road geometries and applied spatial filtering using bounding boxes to ensure topological consistency and data cleanliness
- Developed a Python-based pipeline that converts road shapefiles into an interactive pathfinding network and outputs the optimal route as a GeoJSON file for use in QGIS
- Leveraged graph theory and spatial data analysis to support feasibility studies and autonomous routing for Hyperloop infrastructure proposals

#### QWASE (Queen's Women in Applied Science and Engineering) | Kingston, ON

*Oct. 2023- April 2024*

- Developed and maintained the conference website, ensuring a user-friendly experience for attendees as the marketing coordinator
- Conducted market research to identify trends and opportunities, contributing to a 20% increase in conference attendance
- Created promotional materials, including social media content, email campaigns, and sponsorship packages.

#### Queen's Biomedical Innovation Team | Kingston, ON

*Oct. 2021- April 2022*

- Developed a prototype app using Figma and Qt to process thermography sensor data for breast cancer detection.
- Collaborated with team members to design and implement the app's user interface and functionality.

### PROJECTS

#### NetGen- AI Circuit Design Assistant | SpurHacks | React, Firebase, GPT-4, Python

- Built a web-based assistant that generates LTSpice-compatible netlists from user prompts such as "voltage divider"
- Developed backend logic to parse prompts and output downloadable .net files for immediate LTSpice simulation
- Integrated GPT-4 for natural language processing and automated circuit generation tailored to electrical engineering students

#### Ultron AI Chatbot Web App | HTML, CSS, JavaScript, Flask, OpenAI API

- Designed and developed a full-stack web application styled after *Ultron* from Marvel, featuring dynamic animations and a custom dark-themed UI using Bootstrap and vanilla CSS
- Built an interactive chatbot powered by the OpenAI API, hosted via a Flask backend with CORS and custom prompt engineering for personality emulation
- Implemented frontend logic for typing and randomized text animation to mimic cinematic visual effects and enhance engagement
- Enabled real-time question handling with asynchronous JavaScript and secure API calls through a local Flask server

### REFERENCES

Available on request